

Introduction to structural equation modelling

Alvin Tan
LEVANTE Hackathon | 2026-07-07



Outline

- Structural equation modelling basics
 - Latent vs observed variables
 - Path diagrams
 - Measurement model vs structural model
- Factor analysis
 - Confirmatory vs exploratory factor analysis
 - Using and evaluating factor analytic models
 - Worked example

Structural equation modelling

An intuition

Suppose we wanted to measure intelligence. How would we do so?

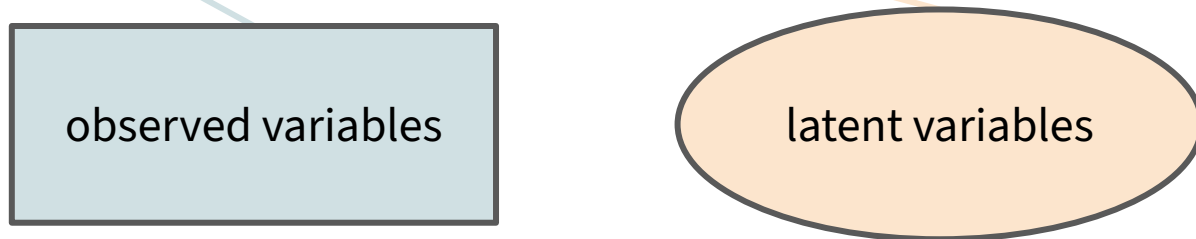
- (e.g., Stanford–Binet, Raven’s progressive matrices, ...)

Let’s say we choose the Stanford–Binet Intelligence Scale. Does it *actually* measure intelligence?

- (Not necessarily—also captures attention, motivation, prior experience with timed assessments...)

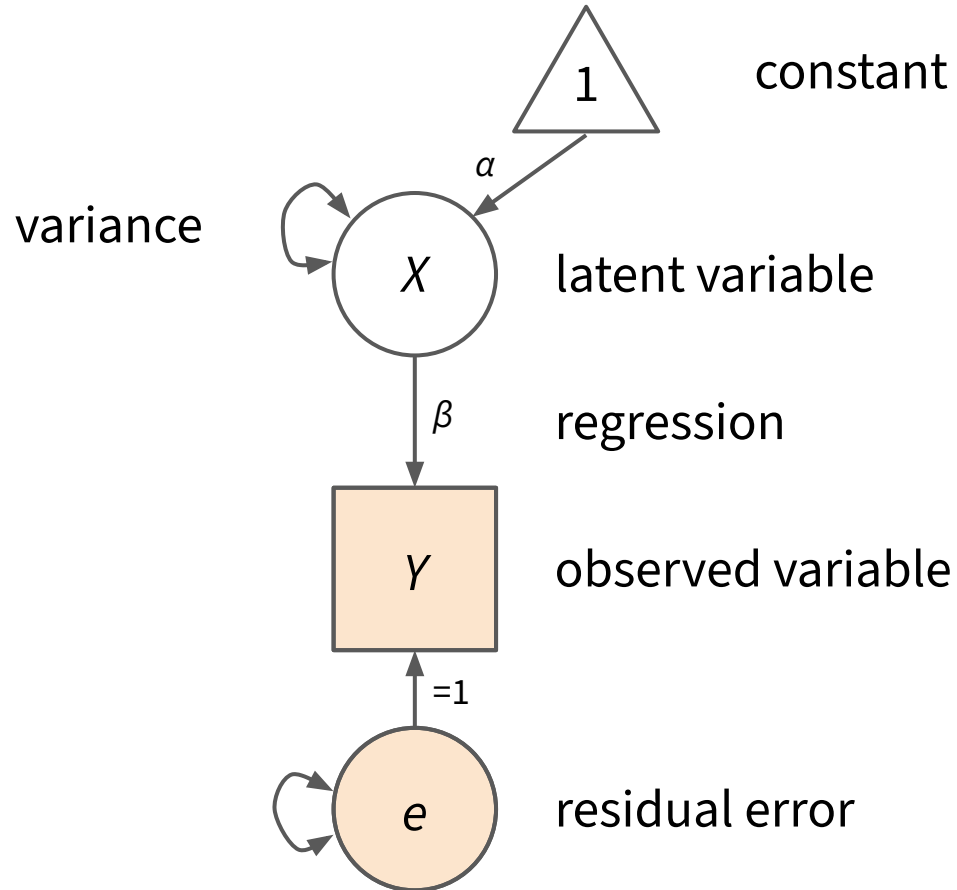
An intuition

In psychology (and in other disciplines), there is a difference between what we can *measure*, and the *constructs* we're actually interested in

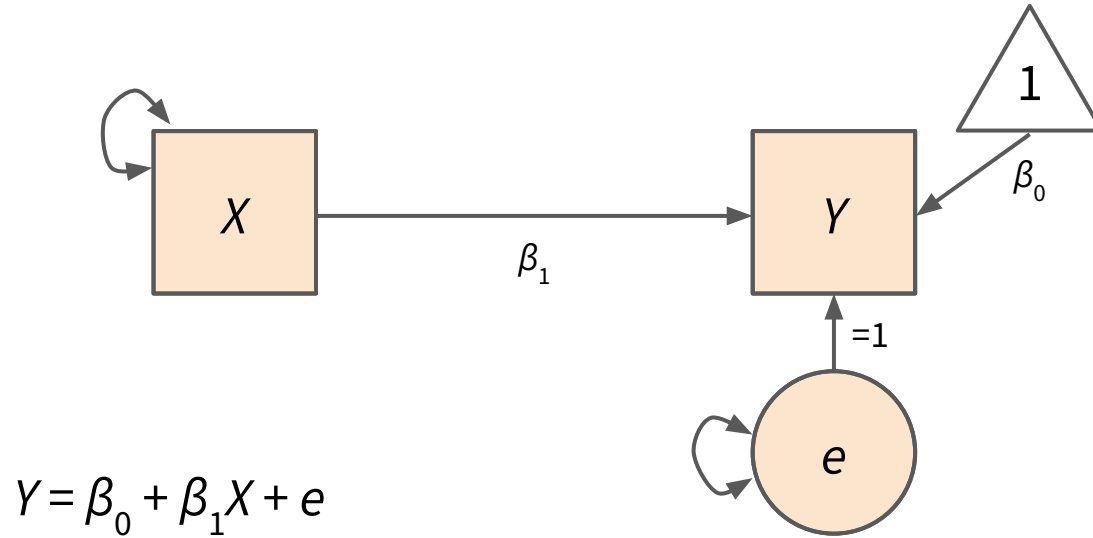


Structural equation modelling is a family of methods that capture the *relationships* among latent and observed variables

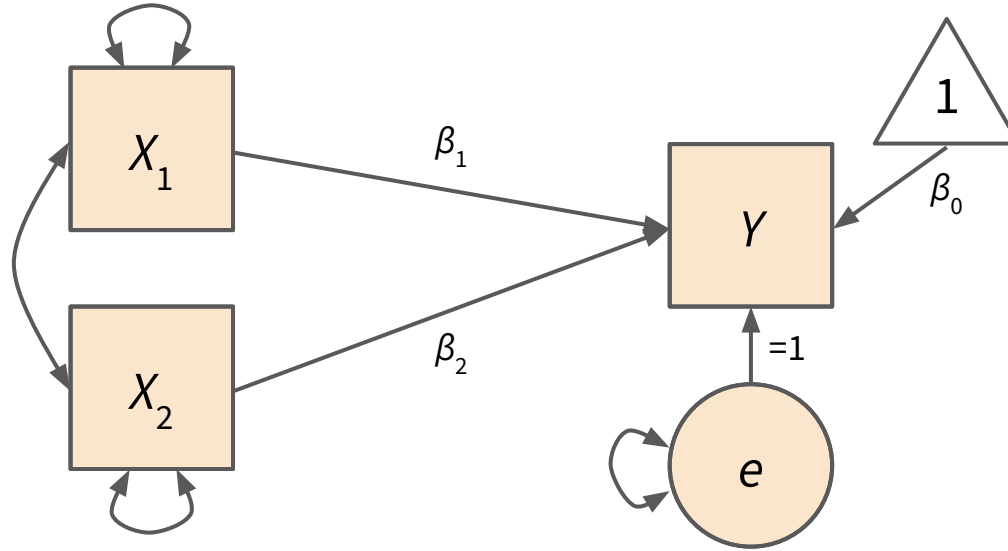
Path diagrams



Path diagrams: Simple regression

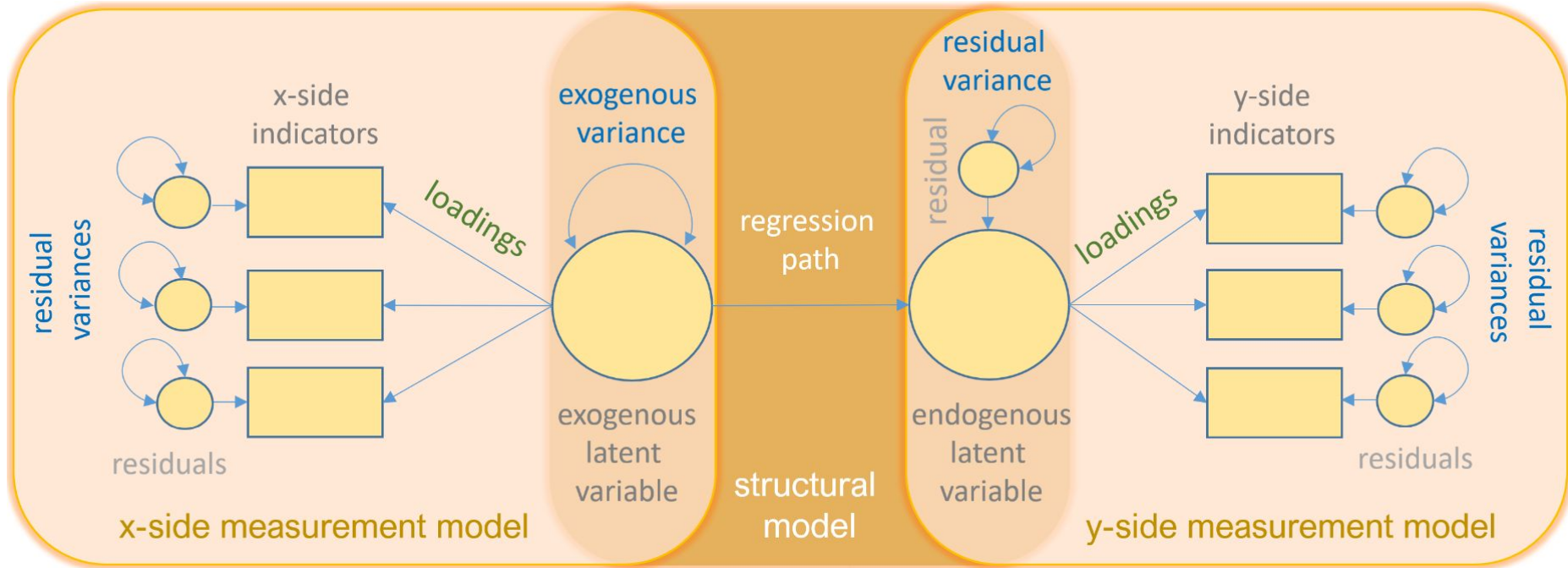


Path diagrams: Multiple regression

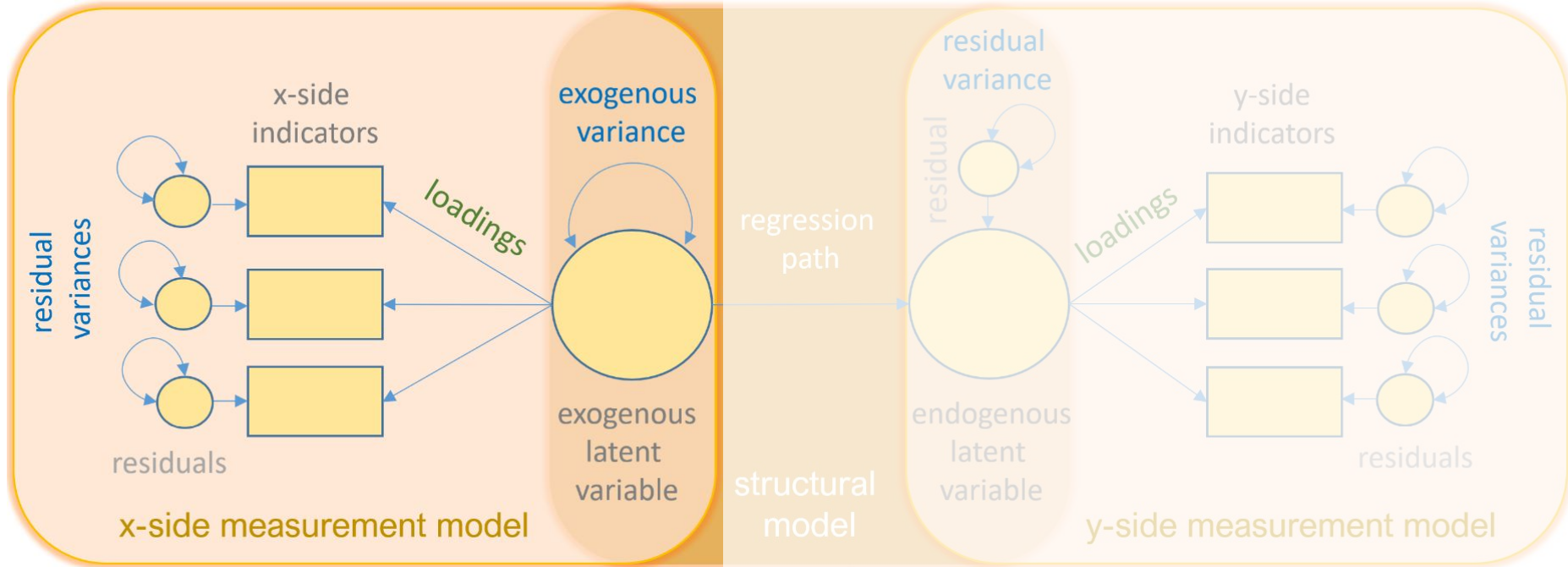


$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + e$$

Path diagrams: Measurement and structural models



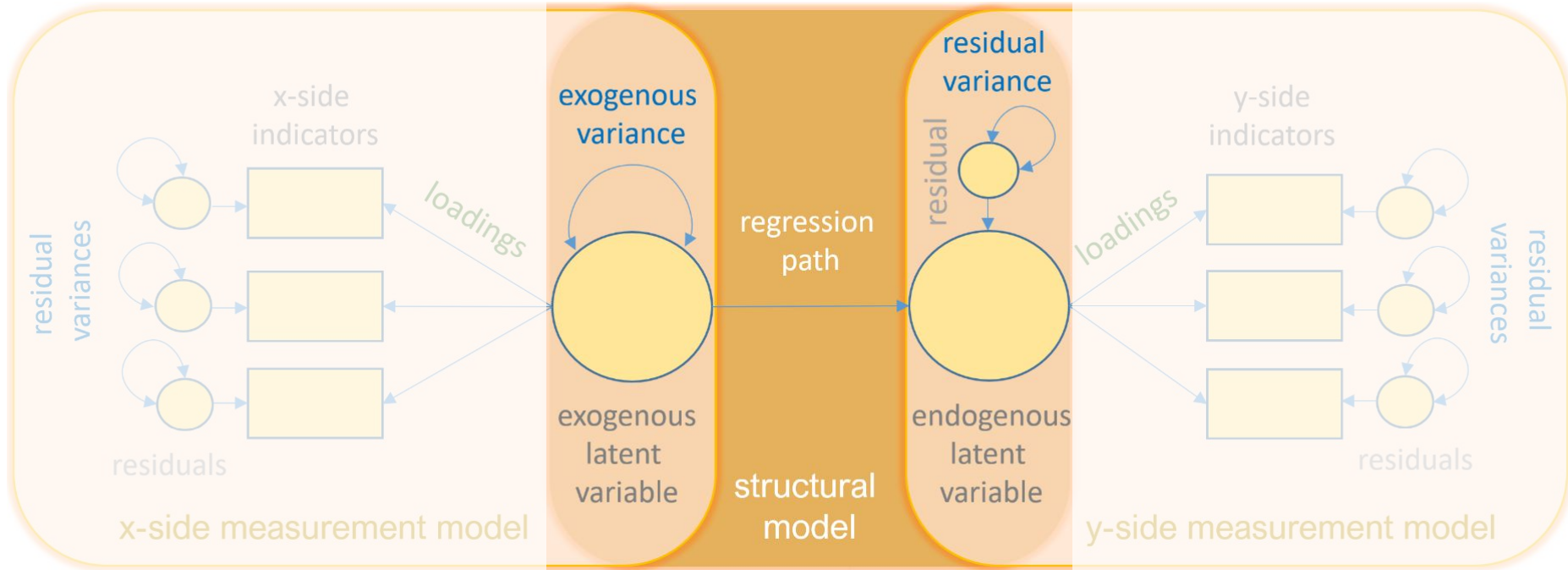
Path diagrams: Measurement and structural models



Measurement model

Links observed & latent variables

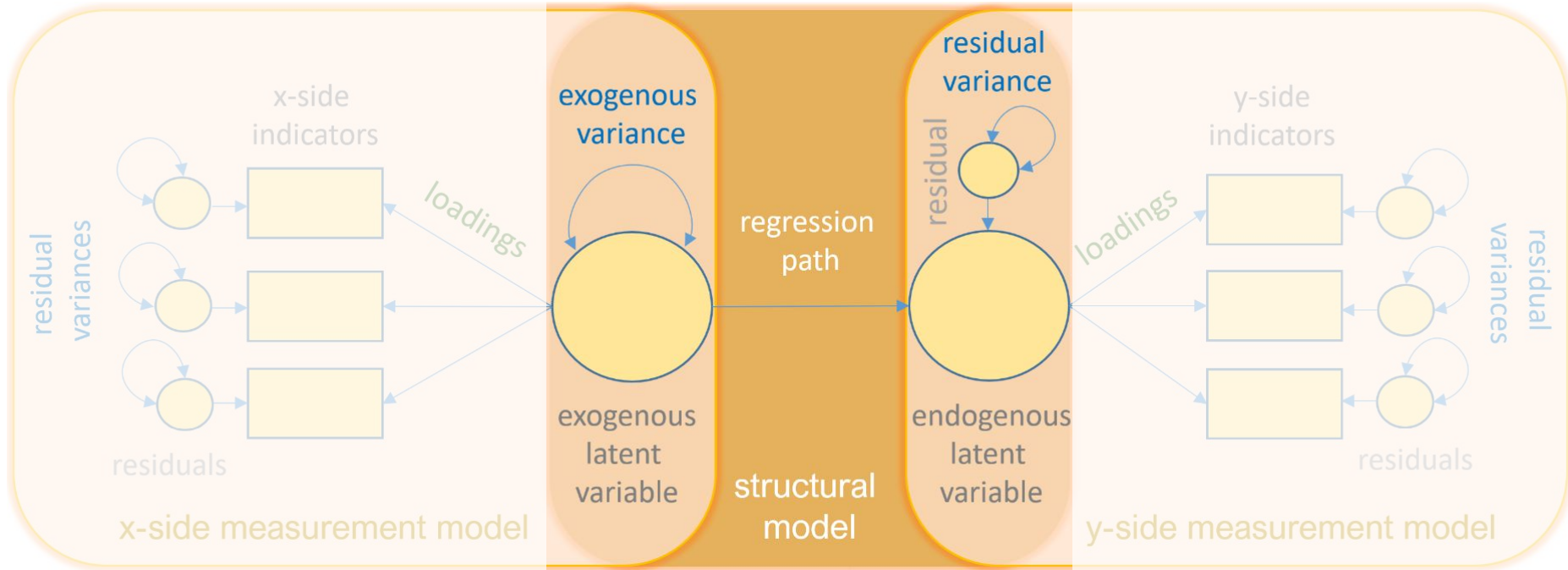
Path diagrams: Measurement and structural models



Structural model

Expresses causal relationships (among latent/observed variables) Lin (2021)

Path diagrams: Measurement and structural models



Exogenous: Independent variable

Endogenous: Dependent variable

Structural equation modelling

SEM (and its extensions) unify many different modelling frameworks, with a focus on explicitly representing and relating latent and observed variables

Some kinds of SEM:

- (Simple/multiple) regression
- Multivariate regression
- Cross-lagged panel models
- Latent growth models
- **Factor analysis**

Factor analysis

Classical test theory

Suppose you had a scale...

Beck Depression Inventory

Name Justine Locke

Date 8 October 2021

Score 26

Big Five Personality Test

Instructions:

- For each statement, select the response that best reflects how you typically feel or behave.
- There are no right or wrong answers. Be honest and choose the response that feels most accurate for you. Please answer all questions.
- 1 = Strongly Agree, 2 = Agree, 3 = Neutral, 4 = Disagree, 5 = Strongly Disagree

Openness to Experience

1. I enjoy trying new things and experiences.

1 2 3 4 5
☒ ☐ ☐ ☐ ☐

2. I am curious about the world and different cultures.

☐ ☒ ☐ ☐ ☐



MacArthur-Bates CDI Words and Sentences

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Proper Mark

USE NO. 2 PENCIL ONLY

Improper Marks

PART I WORDS CHILDREN USE

A. VOCABULARY CHECKLIST

Children understand many more words than they say. We are particularly interested in the words your child SAYS. Please go through the list and mark the words you have heard your child use. If your child uses a different pronunciation of a word (for example, "raffle" instead of "giraffe" or "sketti" for "spaghetti"), mark the word anyway. Remember that this is a "catalogue" of all the words that are used by many different children. Don't worry if your child knows only a few of these right now.

1. SOUND EFFECTS AND ANIMAL SOUNDS (12)

baw baw	☐	meow	☐	uh oh	☐
choo choo	☐	moo	☐	vroom	☐
cockadoodiedoo	☐	ouch	☐	woof woof	☐
grrr	☐	quack quack	☐	yum yum	☐

...how would you determine a test-taker's level?

Classical test theory

One obvious answer: just add it up (= sum scores/true scores)!

$$Y_i = T_i + e_i$$

Y_i : observed score

T_i : true score

e_i : error

With k items,

$$(\sum Y_i)/k = (\sum T_i)/k + \underbrace{(\sum e_i)/k}$$

Expected value = 0

Variance decreases as k increases

Classical test theory

One obvious answer: just add it up (= sum scores/true scores)!

BUT this assumes that **all items behave exactly the same**

I believe that I look ugly.

I feel irritated all the time.

I feel I am a complete failure as a person.

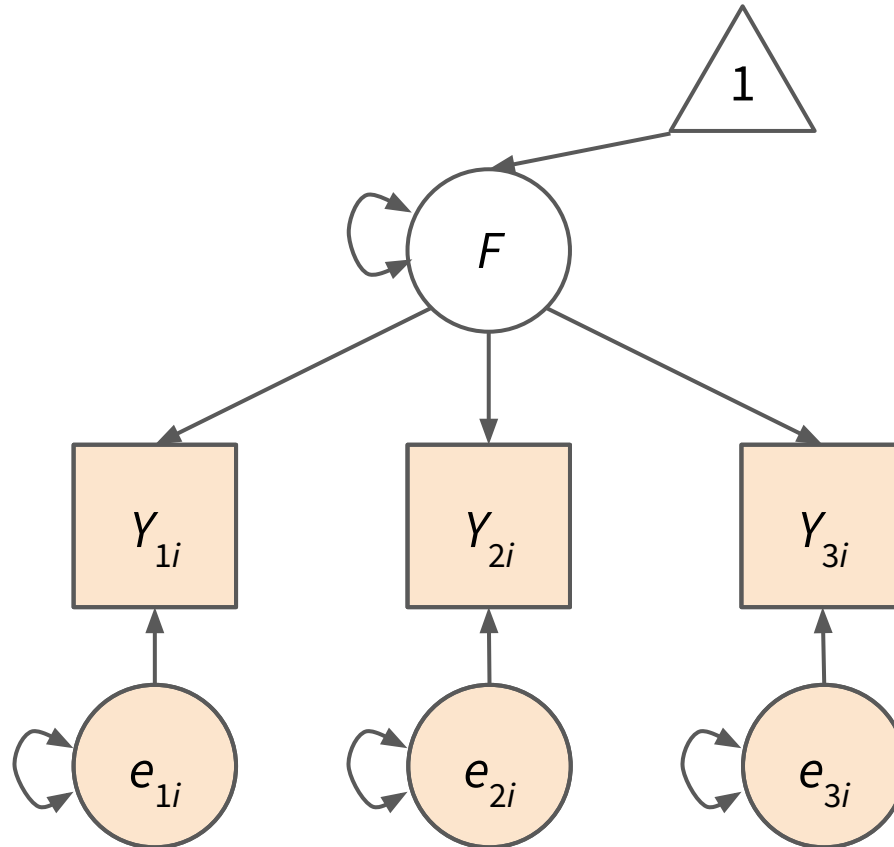
Factor analysis

Factor analysis is a type of *measurement model*: it represents how measured variables may contribute to one or more underlying factors

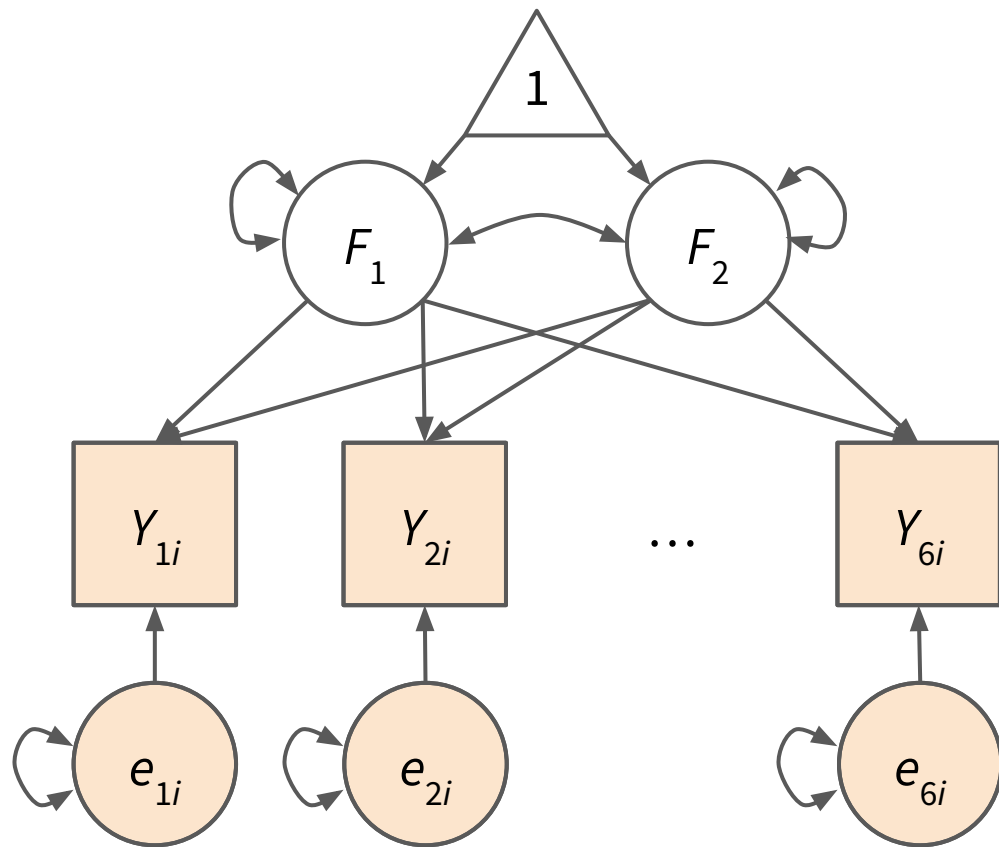
Key ideas:

- Scale items may contribute with different **loadings** (~ strengths) to a factor
- Scale items may be related to **multiple underlying factors**

Path diagram: 1-factor model



Path diagram: 2-factor model



Uses of factor analysis

- Evaluate test validity
- Identify underlying structure
- Test theories about hypothetical constructs
- Reduce data to a smaller number of variables

Two flavours of factor analysis

Exploratory factor analysis (EFA)

- Used to determine the dimensionality of a set of variables
- Data-driven approach (post-hoc examination of structure underlying the data)

Confirmatory factor analysis (CFA)

- Used to study how well a hypothesised structure describes the data
- Hypothesis-driven approach (testing/comparing a priori hypotheses about the data)

Commonly: EFA on an exploratory sample,
then CFA on a confirmatory sample

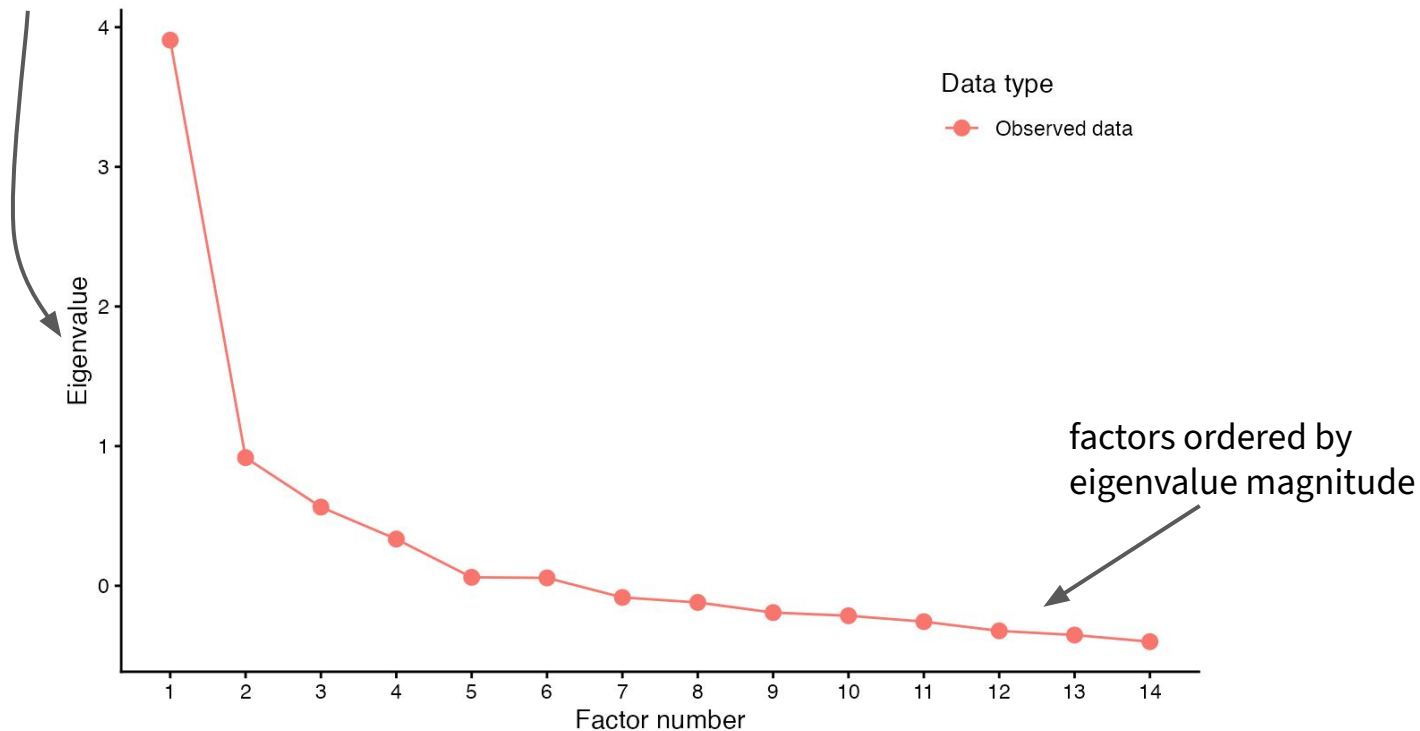
Exploratory factor analysis

1. Select the number of factors
2. Extract the factors + (optionally) rotate factors
3. Calculate factor scores

EFA: Selecting the number of factors

(roughly) how much
variance each factor
explains

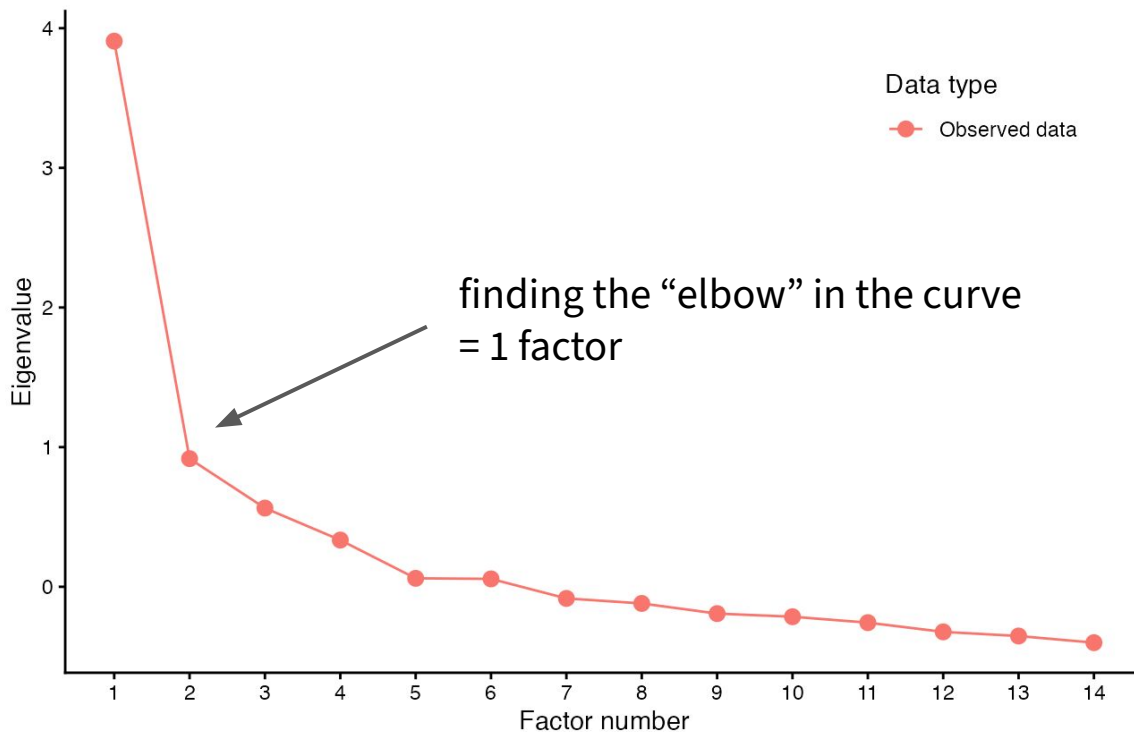
Scree plot



EFA: Selecting the number of factors

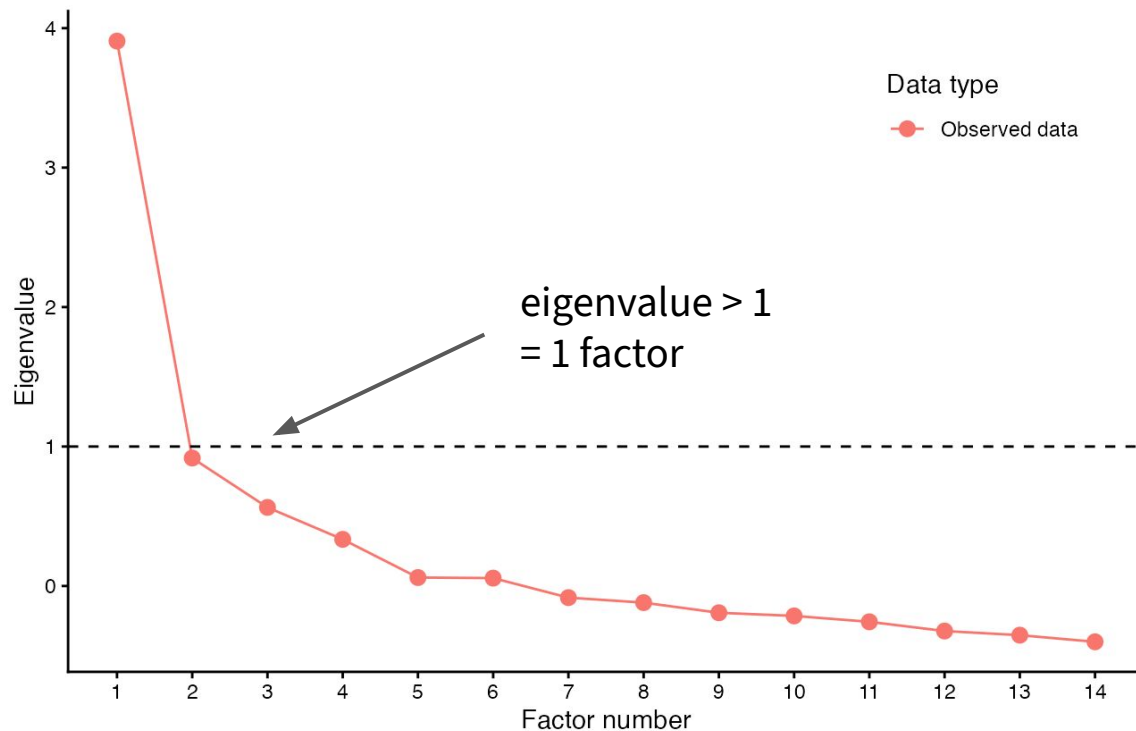
Scree plot

1. Scree test



EFA: Selecting the number of factors

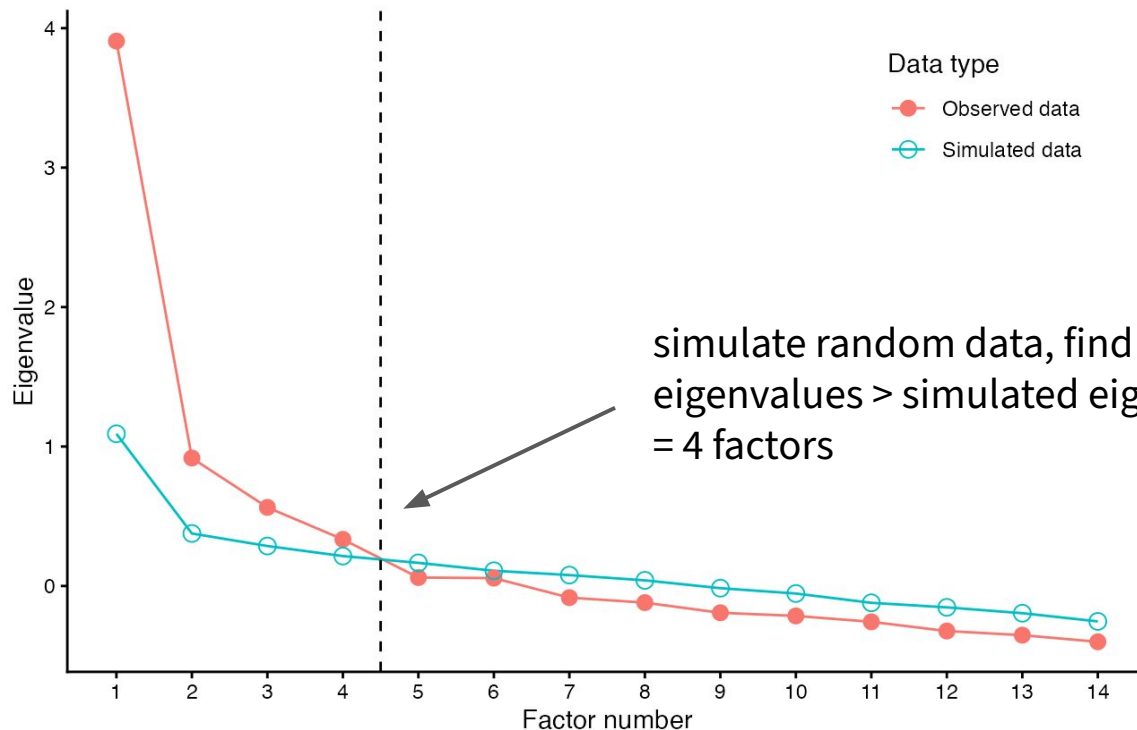
Scree plot



1. Scree test
2. Kaiser criterion

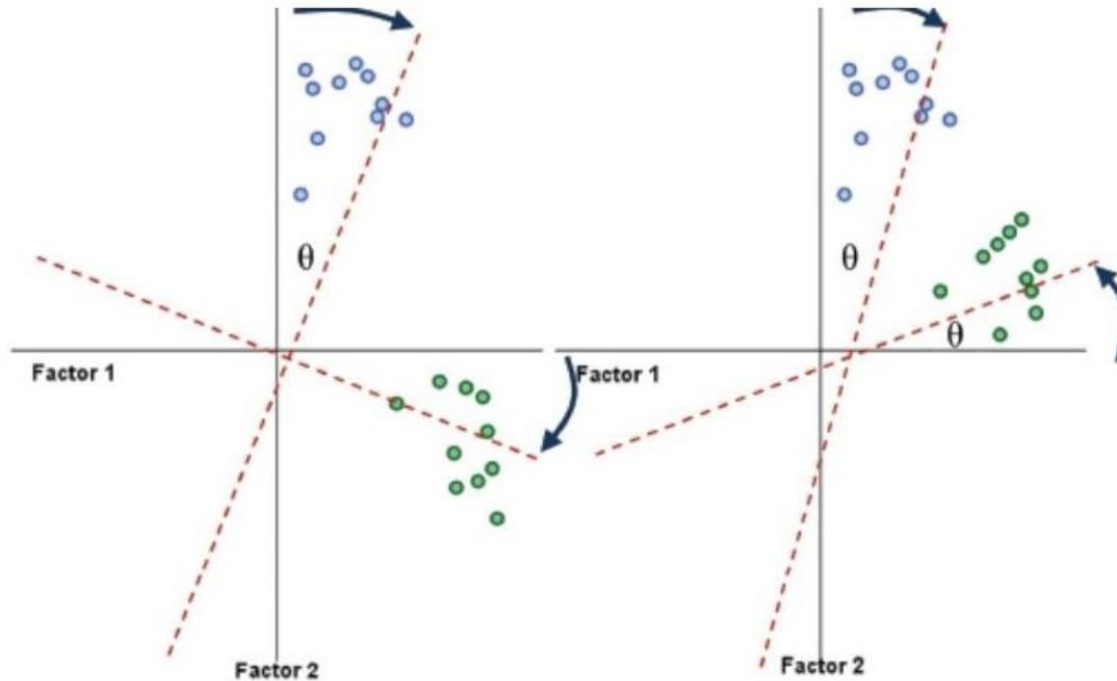
EFA: Selecting the number of factors

Scree plot



1. Scree test
2. Kaiser criterion
3. Parallel analysis

EFA: Extract and rotate factors



Orthogonal Rotation

Oblique Rotation

Rotation separates the factors: items load on a smaller number of factors
→ easier interpretation

EFA: Calculate factor scores

Item	Factor loadings			
	Behavioral management	Freeing parent	Learning	Emotion regulation
The reason for screen use is to make him/her sleepy	0.72	0.11	0.01	0.18
The reason for screen use is to regulate his/her boredom	0.57	0.16	0.22	0.14
The reason for screen use is to help him/her eat better	0.52	0.09	0.14	0.12
The reason for screen use is to spend time with my friends, family, or partner (e.g., to have a quiet meal)	0.49	0.09	0.15	0.32
The reason for screen use is to get my housework done	0.14	0.69	0.20	0.04
The reason for screen use is to get work done	0.58	0.67	0.09	-0.07
The reason for screen use is to calm myself down, when I am stressed	0.09	0.65	0.26	0.35
The reason for screen use is to have some peace for myself	0.14	0.52	0.31	0.36
The reason for screen use is to learn something new	0.19	0.11	0.80	-0.10
The reason for screen use is to improve his/her native language	0.24	0.18	0.57	-0.03
The reason for screen use is to learn a foreign language	0.13	0.17	0.54	0.09
The reason for screen use is to be prepared for the digital future	-0.05	0.11	0.46	0.18
The reason for screen use is to calm him/her down	0.35	0.22	0.18	0.50
The reason for screen use is to help him/her to not be angry anymore	0.23	0.07	-0.05	0.44

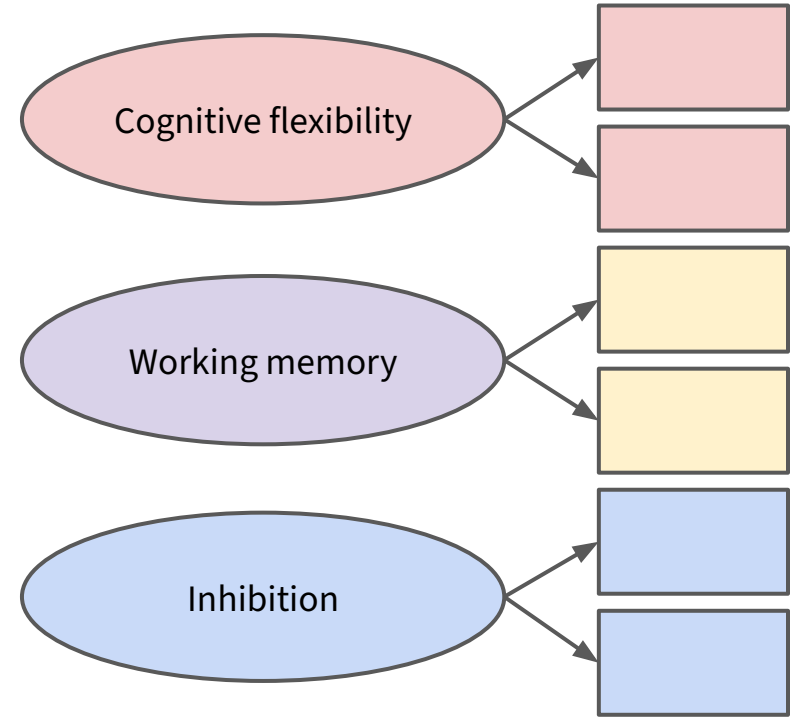
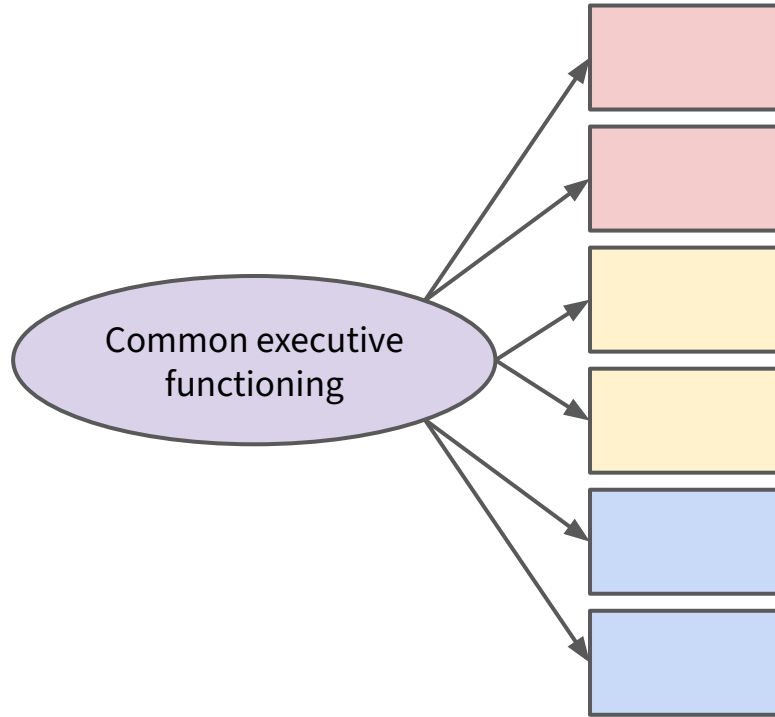
Exploratory factor analysis

1. Select the number of factors
2. Extract the factors + (optionally) rotate factors
3. Calculate factor scores

Confirmatory factor analysis

1. Construct hypothesised factor structure(s)
2. Estimate parameters
3. Evaluate model

CFA: Construct hypothesised factor structure



CFA: Estimate parameters



CFA: Evaluate model

- Chi-square
 - Deviation between observed and model-predicted covariances; smaller = better
- Root mean square error of approximation
 - Covariance deviation adjusting for sample size; smaller = better
- Incremental fit indices
 - E.g., comparative fit index (CFI), normed fit index (NFI), Tucker–Lewis index (TLI)
 - Model deviation compared against baseline model deviation; larger = better
- Akaike / Bayesian information criteria (AIC/BIC)
 - Only for model comparison
 - Model log likelihood adjusting for number of parameters; smaller = better

Confirmatory factor analysis

1. Construct hypothesised factor structure(s)
2. Estimate parameters
3. Evaluate model

Factor analysis: Assumptions

- Continuous measures
- (Approximately) multivariate normal distribution
- Reasonably large number of observations
- Independent observations

Worked example

Executive Function from Observation and Reflection Tool

(from the LEVANTE Caregiver Survey – Child)

Where to?

- Investigate underlying structure of data
- Explore causal networks among latent variables
- Model longitudinal change over time
- Extend to nonlinear relationships
- Test for multi-group variability in data structure (tomorrow!)
- ...and much more!

ty+q?